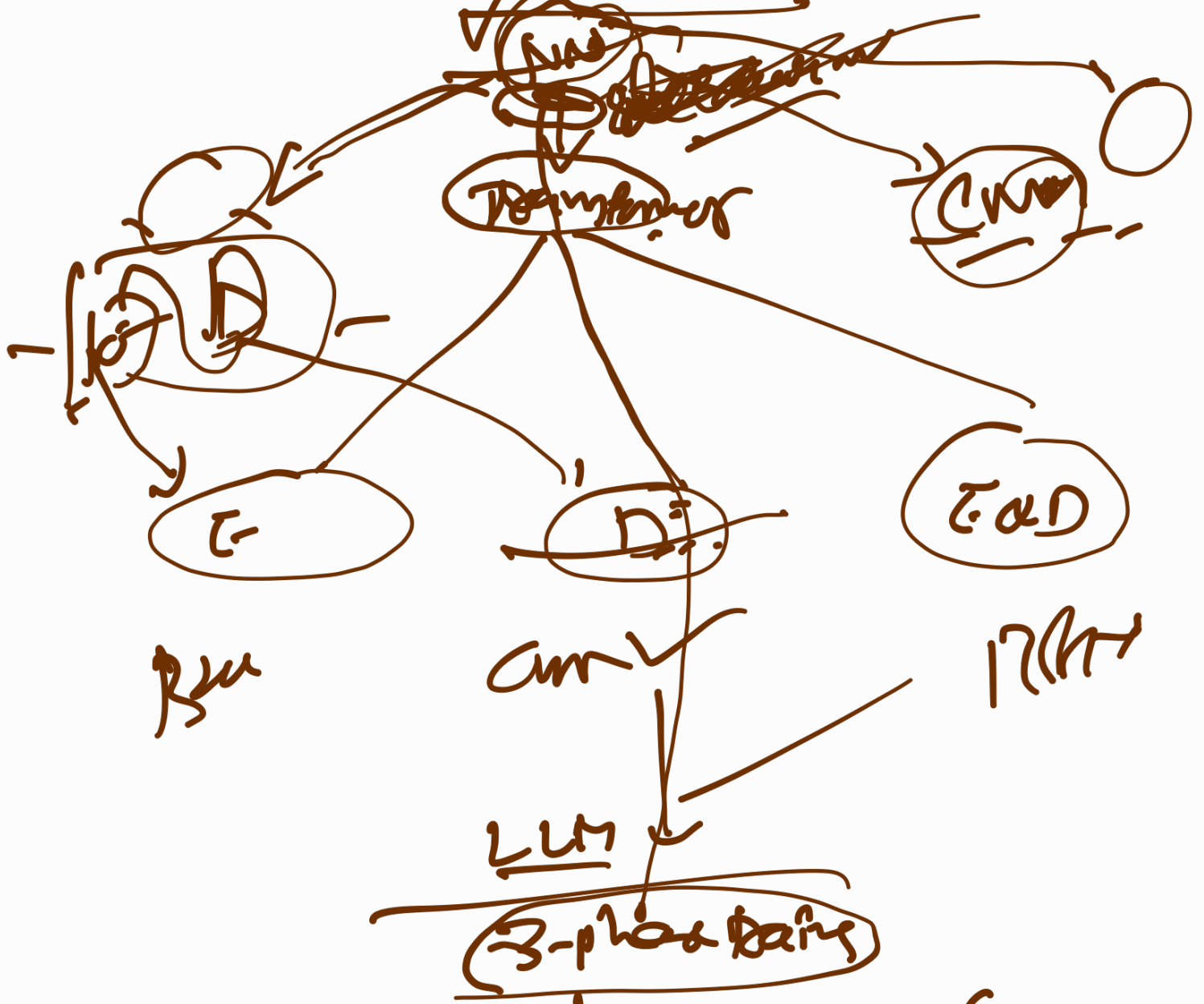
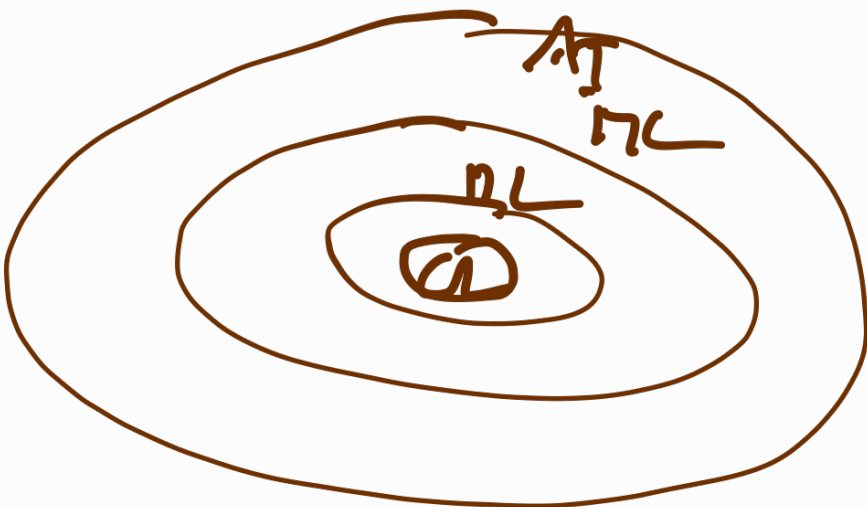


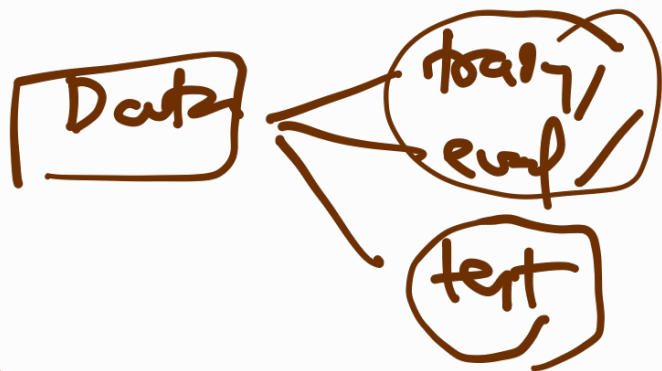
overview of AI



→ pattern learning without explicit programming ✓

AI → ML → DL → GL





teaching



answer

dev

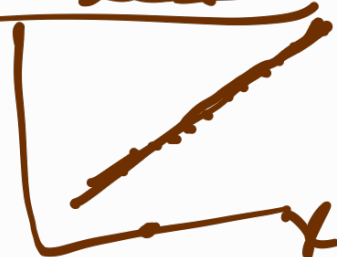
00

Linear function learn ^{single variable}

test

ku

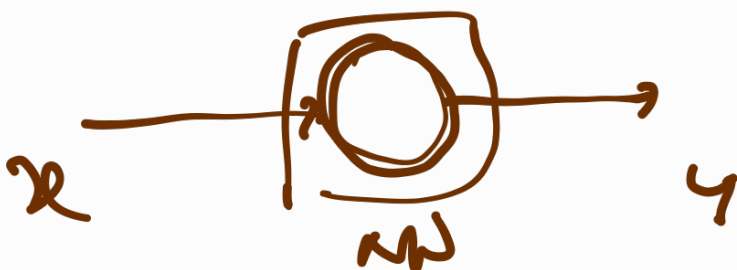
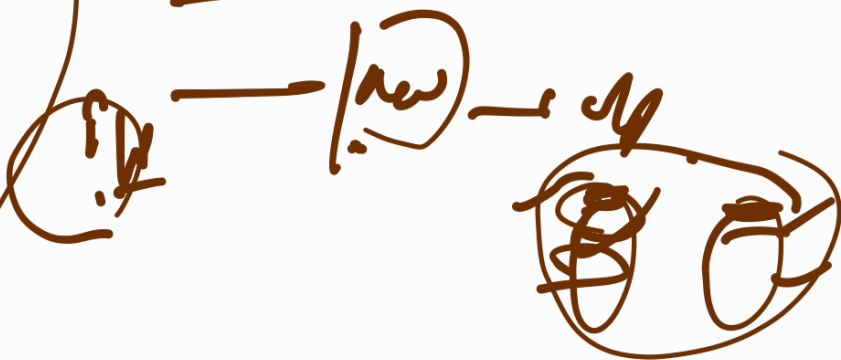
✓



$$y = 2x + 1 + \text{noise}$$



eva



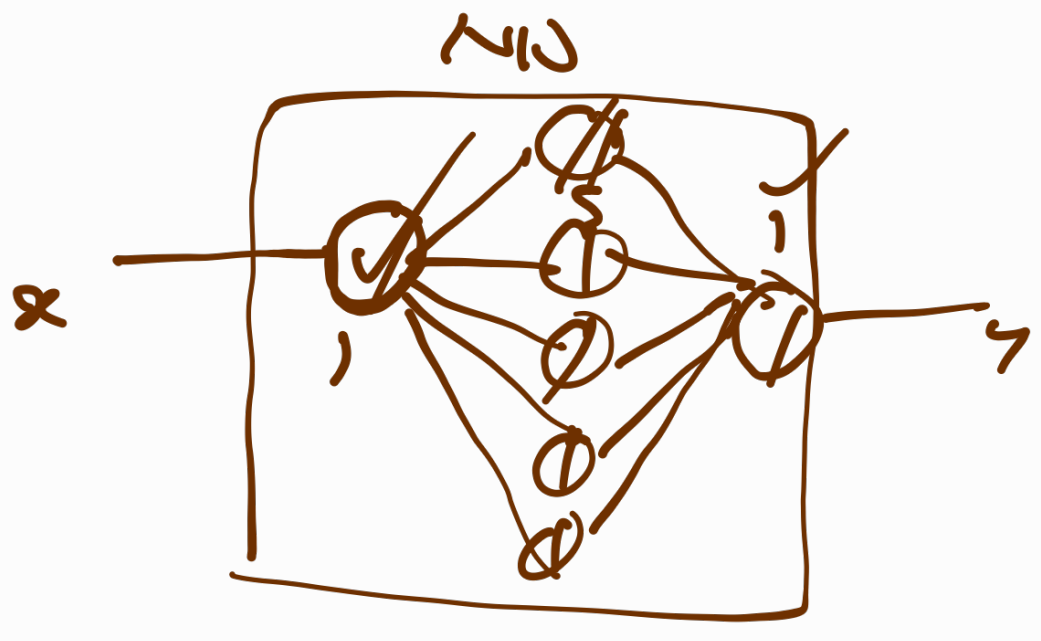
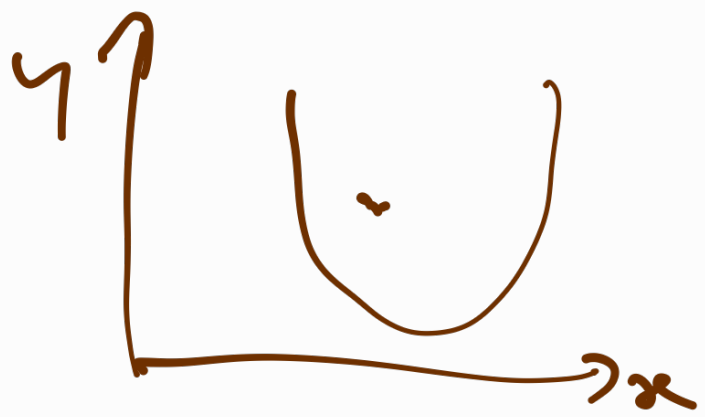
2
1
2
3

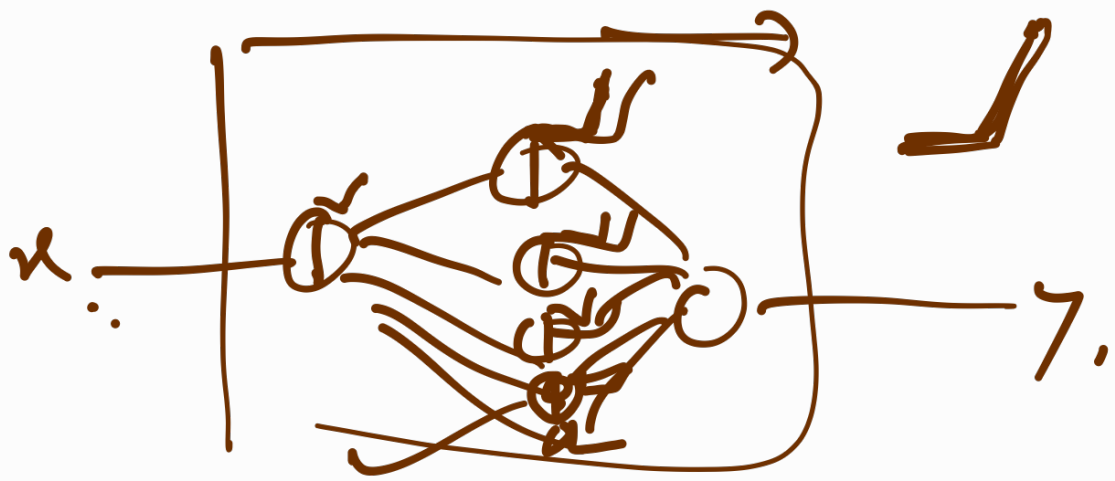
y = 2x + 1 + noise

0.1 + 1 + noise = 1.3

-0.3 + 1 + noise = -

Randomness Function (Stochastizität)



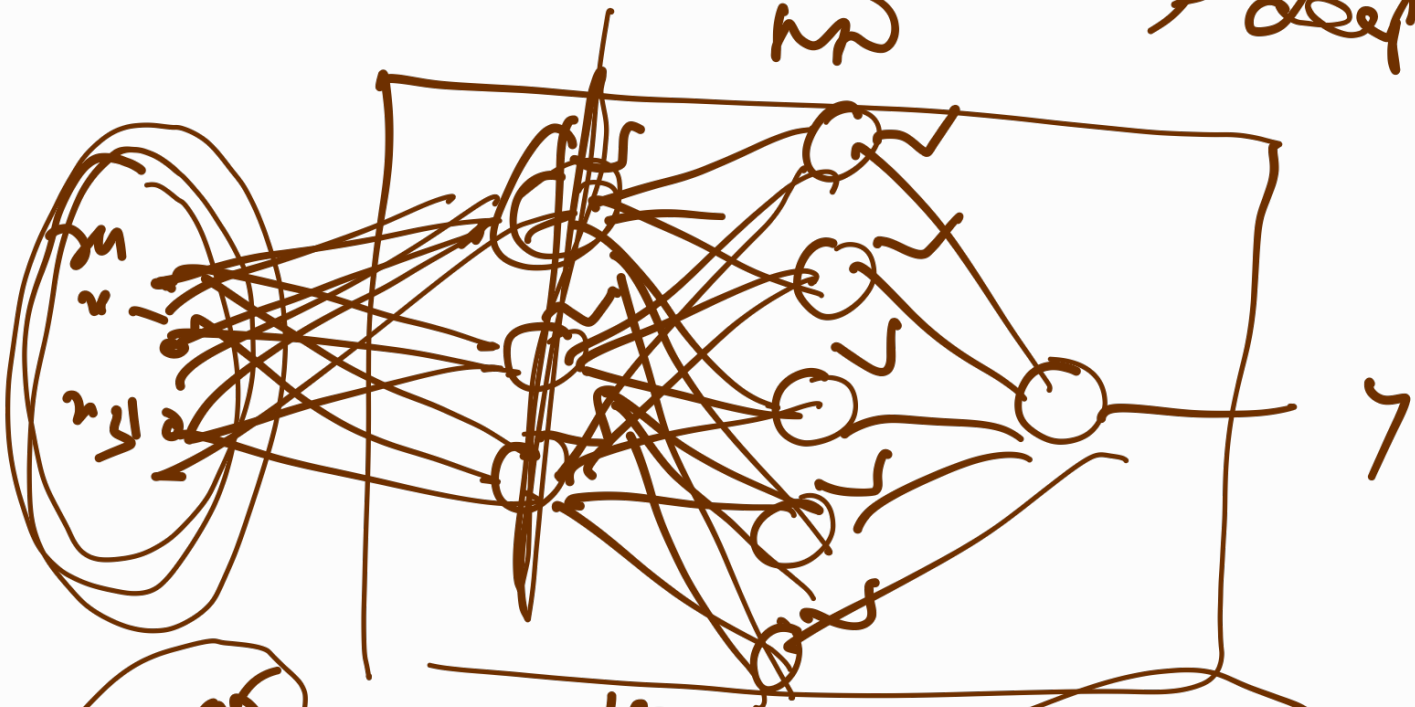


(145) Complex nonlinear function

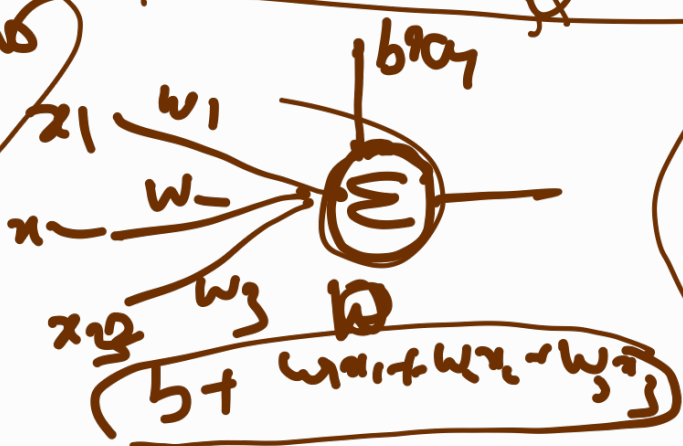
$$y = x_1^2 + \sin(x_2) - e^{x_3} + \dots$$



neuron size
 MN → deep



Linear
 function





$$op = f(b + w_1x_1 + w_2x_2 + w_3x_3)$$

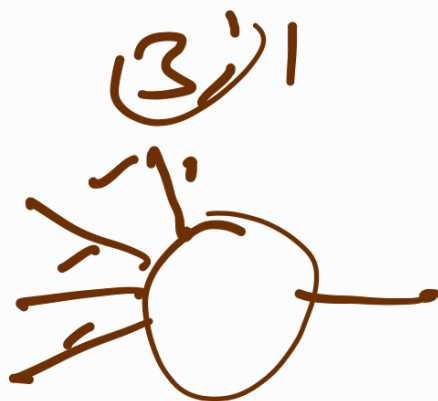


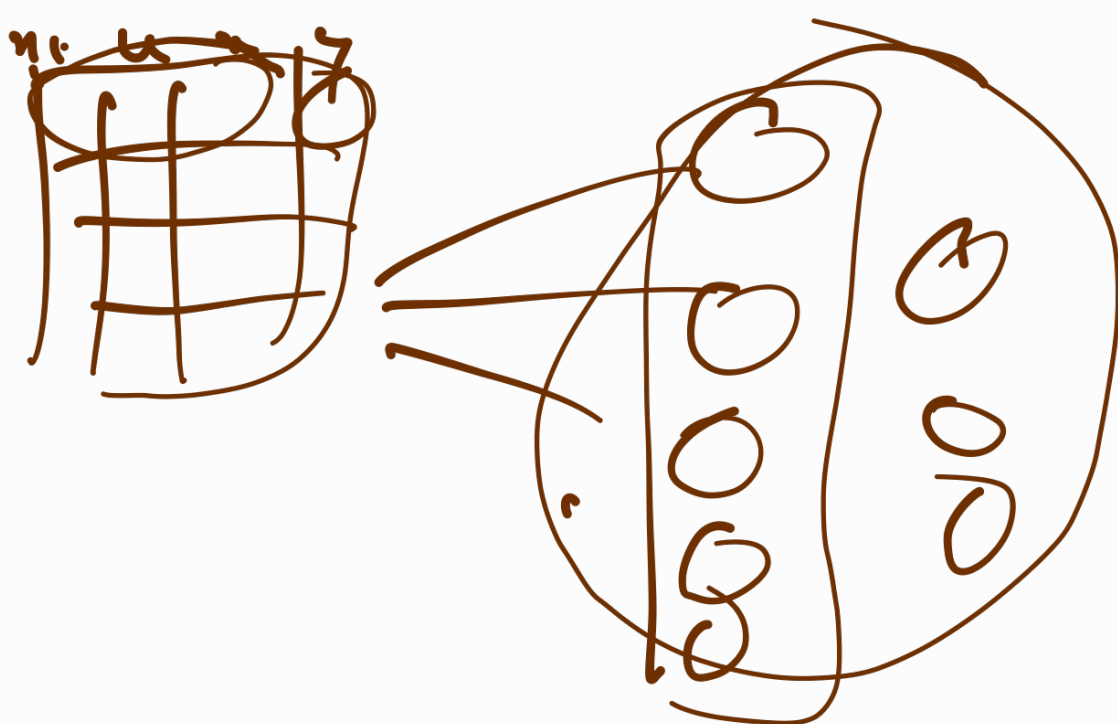
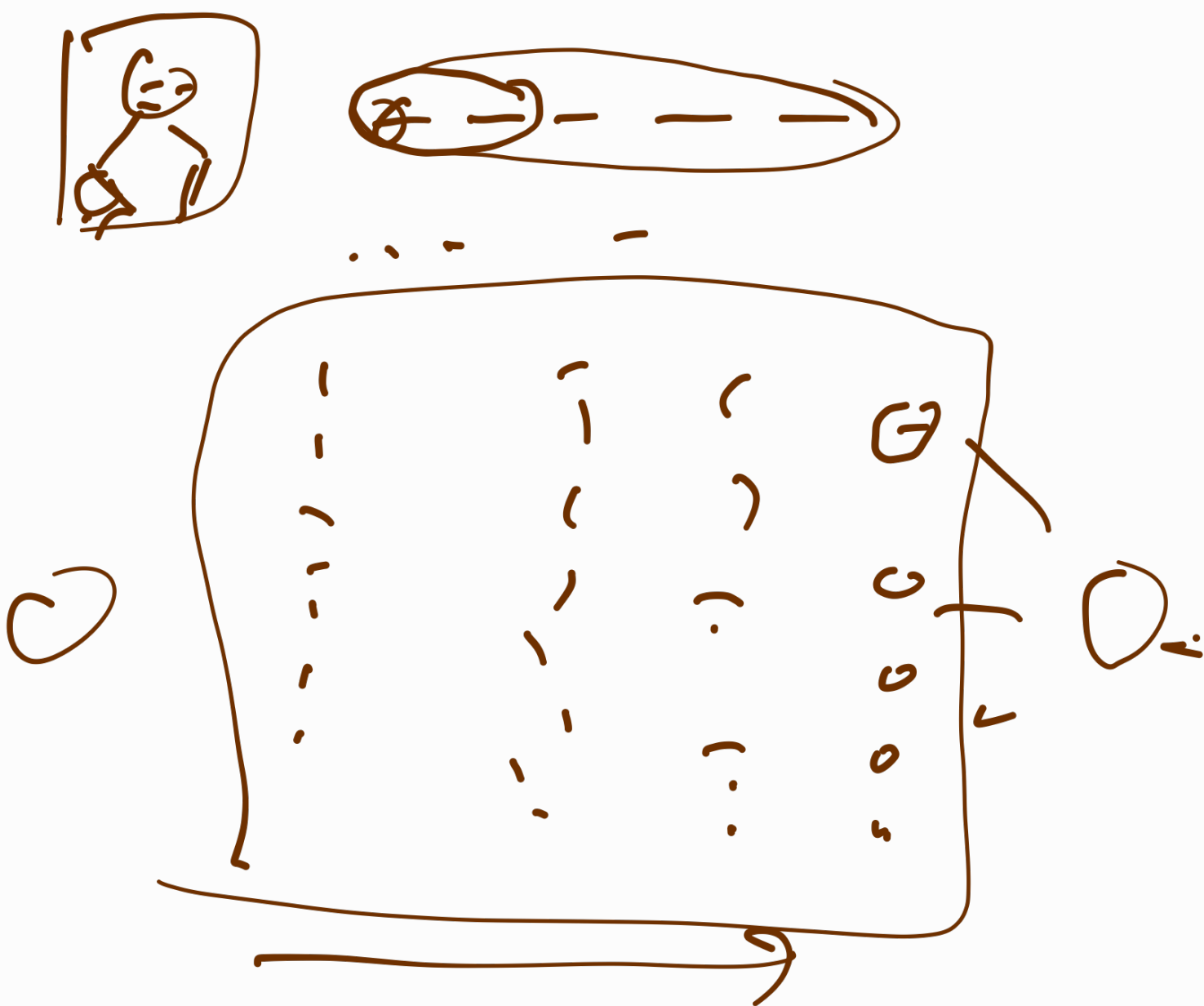
w_0, w_1, w_2, w_3
Weights

Neuron



Neuron

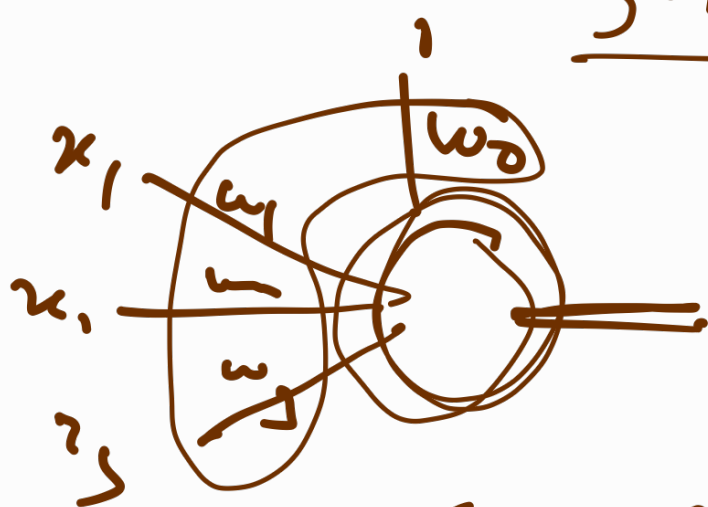




~~(1, 3)~~

~~(3, 1)~~

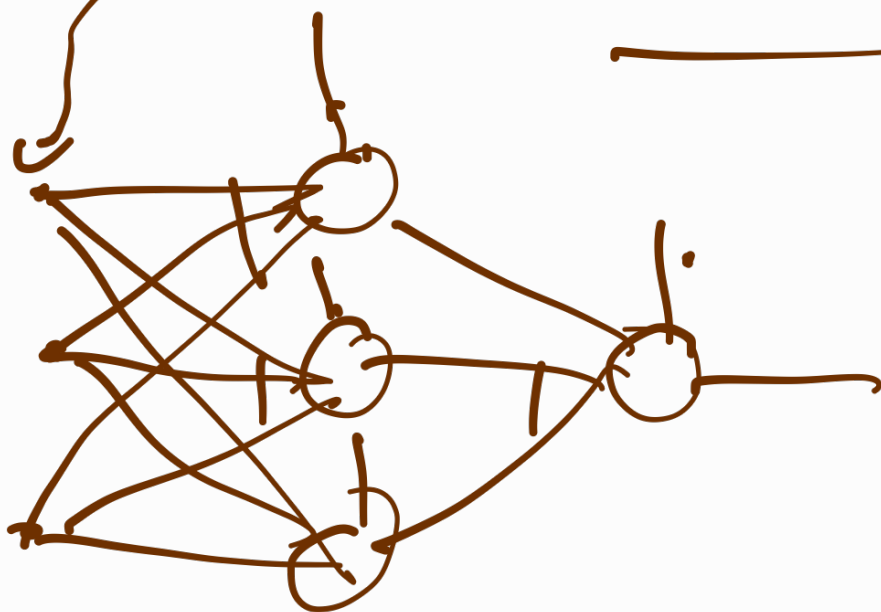
Sort Jan



← weights,

~~(3, 3, 1)~~

2 - Greynwh

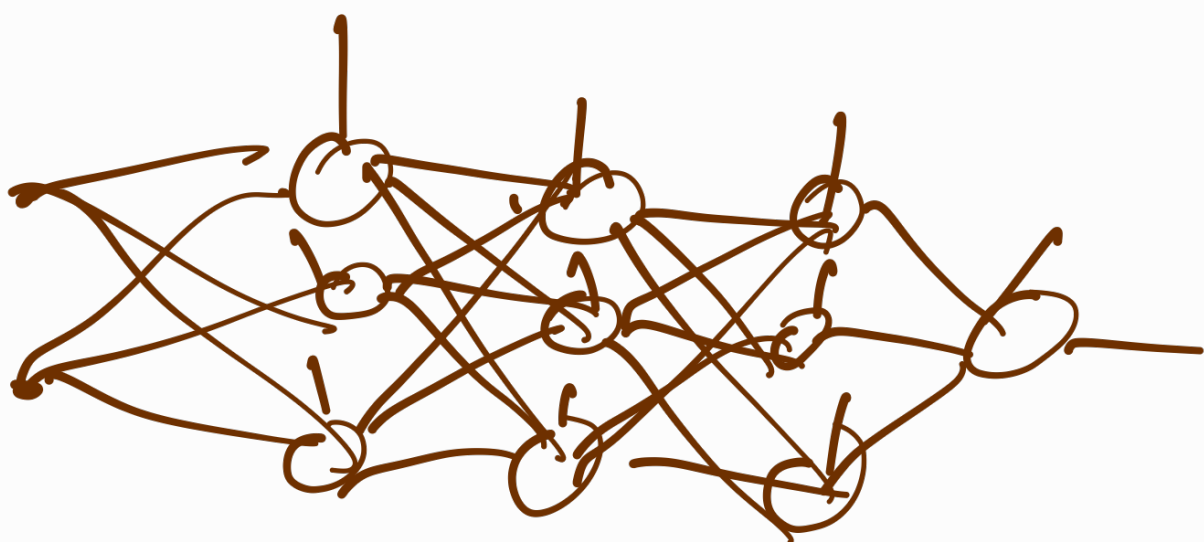


4 3 2 1

4

(6) us

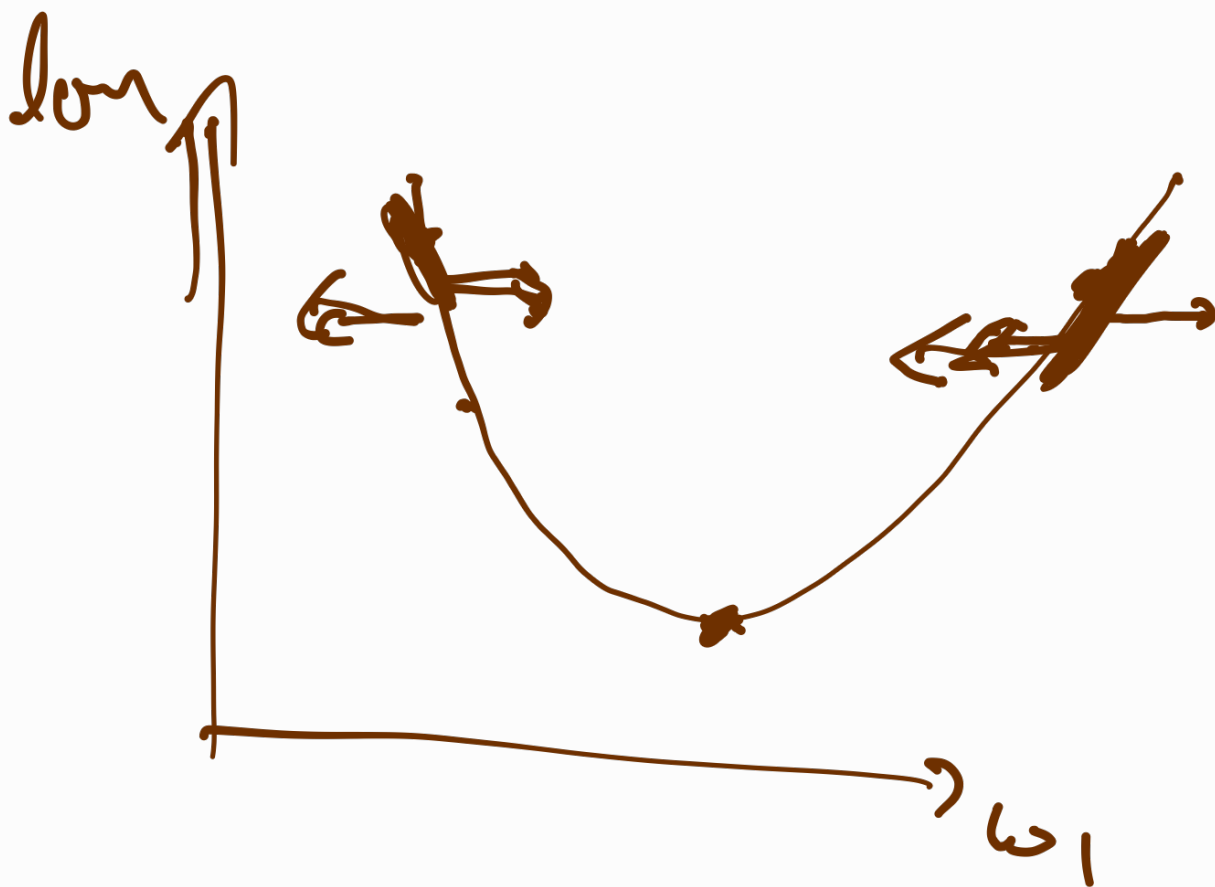
$\{2, 3, 3, 3, 1\}$



$3+5=9$ $4+5+1+2+6+1=19$

37

c





$$\eta = \mu \# \gamma$$

